

What is it about teaching that influences students' learning?

In their literature review about the effects of classroom mathematics teaching on student learning, James Hiebert and Douglas Grouws (2007) pose the question: "What is it, exactly, about teaching that influences students' learning?" To examine this question, they analyzed research studies of classroom teaching to identify the kinds of teaching that are empirically linked to two critical components of mathematical proficiency: skill efficiency and conceptual understanding.

The authors' findings point to several key features that facilitate skill efficiency (fast and accurate execution of mathematical procedures). Evidence suggests that skills and procedures are best taught in a fast-paced setting that includes teacher modeling, teacher-directed questions that require a specific correct answer, and substantial time for students to practice.



Pinning down the answer to what influences students' conceptual development was a more complex task, in part, because of the difficulty of measuring conceptual understanding and disentangling the interactive components of a complex classroom environment. For this reason, Hiebert and Grouws only examined studies where the teaching was described in some detail, enabling them to look

carefully at what was really happening during a particular lesson.

In the multiple studies examined by Hiebert and Grouws, evidence revealed that conceptual understanding occurs in a wide range of teaching methodologies and contexts. Some classrooms were characterized as teacher-centered and others student-centered. Some lessons involved group work; others did not. Some relied heavily on teacher demonstrations, while others highlighted class discussions. In other words, the answer to the question, "What is it about teaching that influences student learning?" cannot be answered by pointing to a single best pedagogical approach. However, Hiebert and Grouws found that evidence does suggest a strong link between student conceptual understanding and two key features of teaching—features that can be present in a variety of

methodologies and settings. These features are making mathematical relationships explicit and allowing students to struggle with important mathematics.

Making Mathematical Relationships Explicit

What does it mean to make mathematical relationships explicit? Hiebert and Grouws explain that this type of teaching treats mathematical relationships between ideas, facts, and procedures in an explicit and public way. It includes such things as reminding students of the main point of the lesson and how it connects to previous and future learning, discussing the mathematical meaning that underlies a procedure, questioning students about how solution strategies are similar to or different from each other, or discussing the ways in which mathematical concepts build on each other. The process of making mathematical relationships explicit can occur in a variety of contexts. Conceptual knowledge can be developed through highly structured teacher-centered instruction. It can also occur within looser, more student-centered approaches. In other words, what matters is the regular opportunity for teachers and students to pay attention to—in an intentional and public way—the mathematical relationships between facts, ideas, and procedures.

Allowing Students to Struggle with Important Mathematics

Evidence also suggests a strong link between student understanding and another feature of teaching—the engagement of students in struggling with important mathematical ideas. Hiebert and Grouws use the word struggle to mean that students are required to think hard to make sense of mathematics and to expend effort wrestling with something that is not immediately apparent. This type of struggle occurs when, for example, students are asked to solve problems that are within reach but require them to grapple with key ideas. Struggle does not mean that students are given tasks that are nonsensical or overly difficult. It does not imply a level of challenge that leads to despair and frustration. Rather, struggle, as used in the context of promoting student learning, must be



designed as a doable challenge where students are clear about what they need to figure out and can proceed with some confidence that they have the tools to help them succeed.

Struggle allows confusion and doubt to surface, setting the stage for students to restructure their mental processes and make sense of the mathematical ideas in a new way. Teaching that successfully incorporates student struggle by using struggle time—whether it be just 5–10 minutes or a more significant portion of the class period—gives students the opportunity to fit ideas together to make sense of a challenging problem or idea. In addition, struggle connects back to the first feature—making mathematical relationships explicit—by mentally preparing students to engage in meaningful discussions and generate important questions about key mathematical ideas.

Interestingly, Hiebert and Grouws also noted that in addition to developing conceptual understanding, teaching that attended to explicit development of mathematical relationships and allowed students to wrestle with important mathematics also facilitated procedural fluency—skill efficiency plus the ability to use skills flexibly and with meaning.